

HAEM-L1 Companion Figures

Anaemia — Phase Ib Haematology Tutorial

21 figures

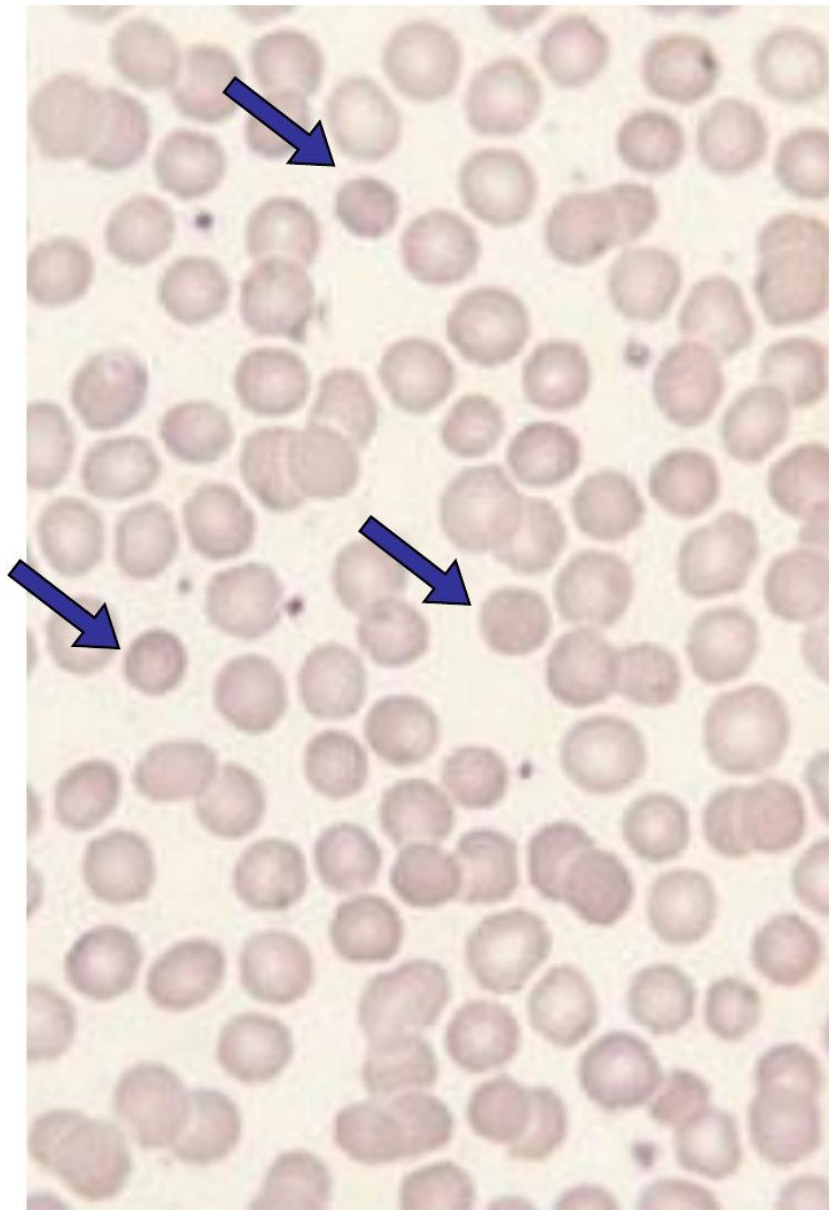


Figure HAEM-L1.01

HS blood film: spherocytes (round cells lacking central pallor)

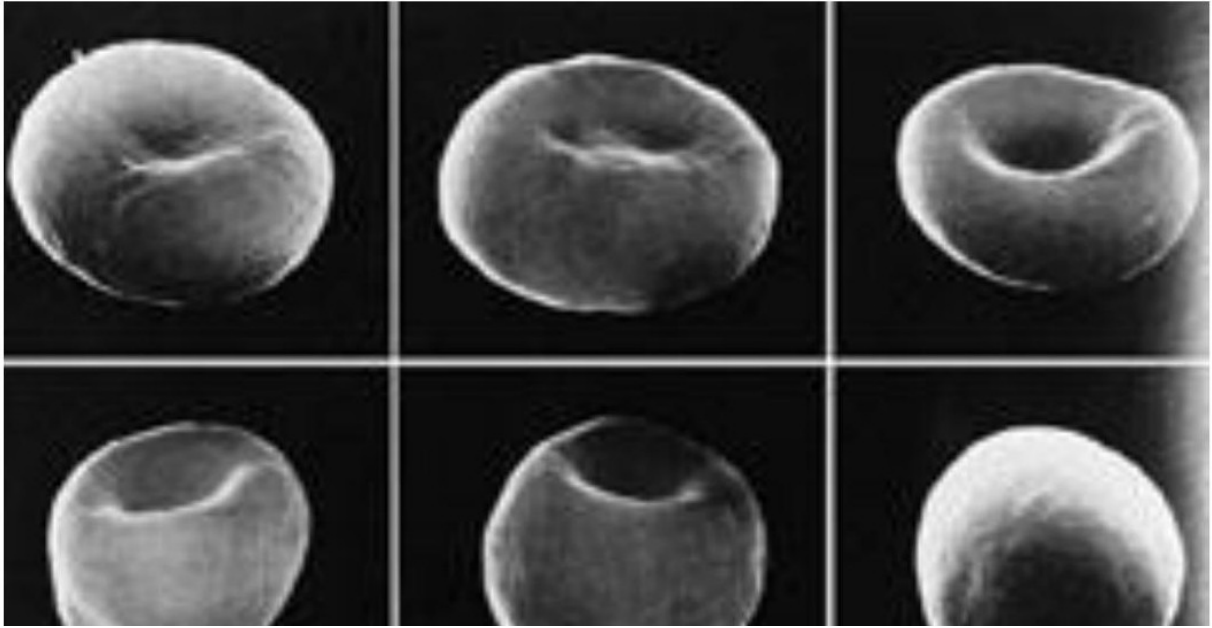
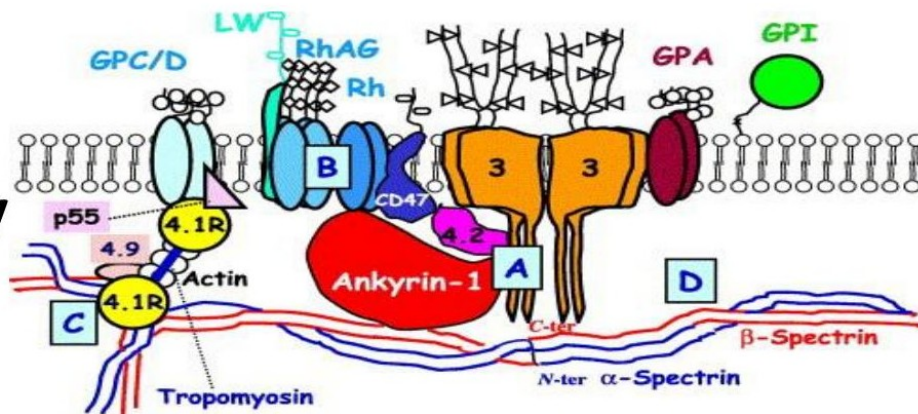


Figure HAEM-L1.02

SEM: spherocyte (sphere) vs normal RBC (biconcave disc)

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logical disorders.

Figure HAEM-L1.03

Red cell membrane skeleton: ankyrin/spectrin vertical linkages

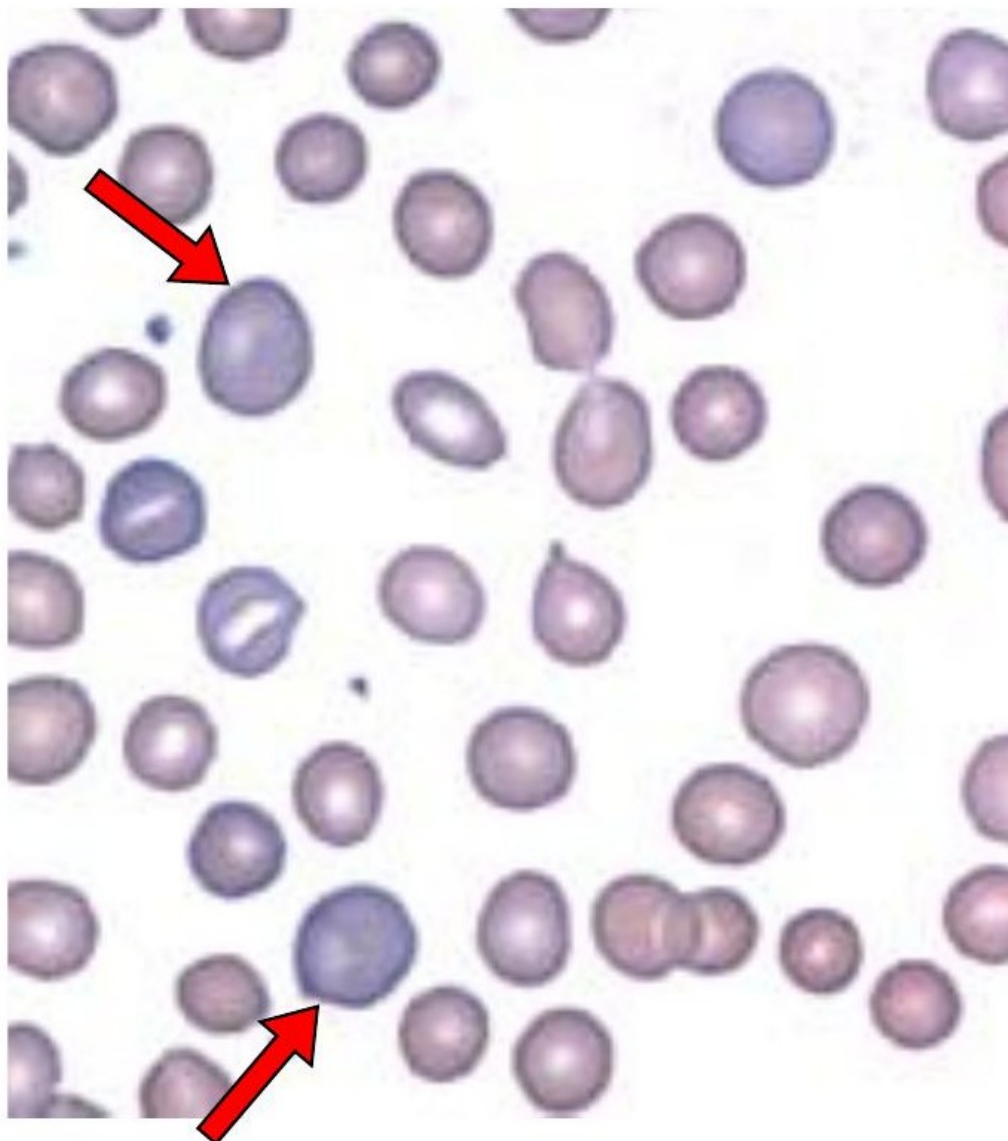


Figure HAEM-L1.04

Polychromatic macrocytes (reticulocytes, arrowed) — marrow response to haemolysis

b) Case 1

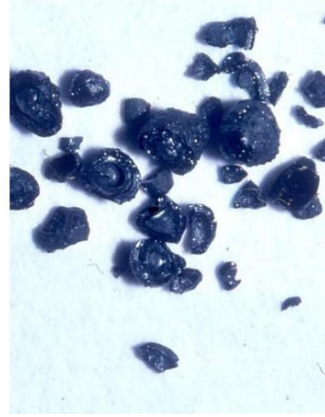


Figure HAEM-L1.05

Pigment gallstones in chronic haemolysis (cholangiogram + extracted stones)

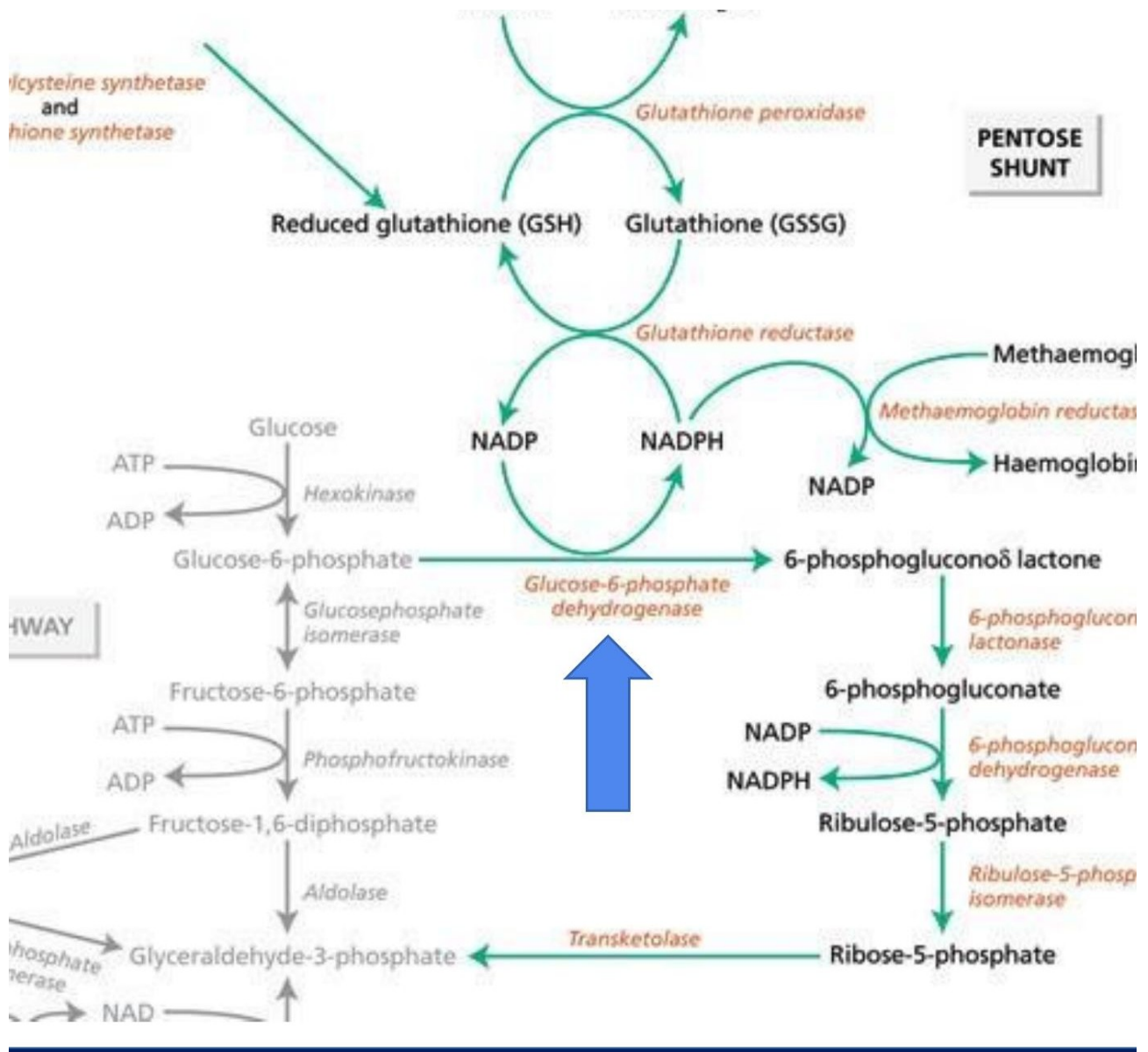


Figure HAEM-L1.07

Glycolysis and pentose phosphate (HMP) shunt; G6PD generates NADPH for glutathione cycle

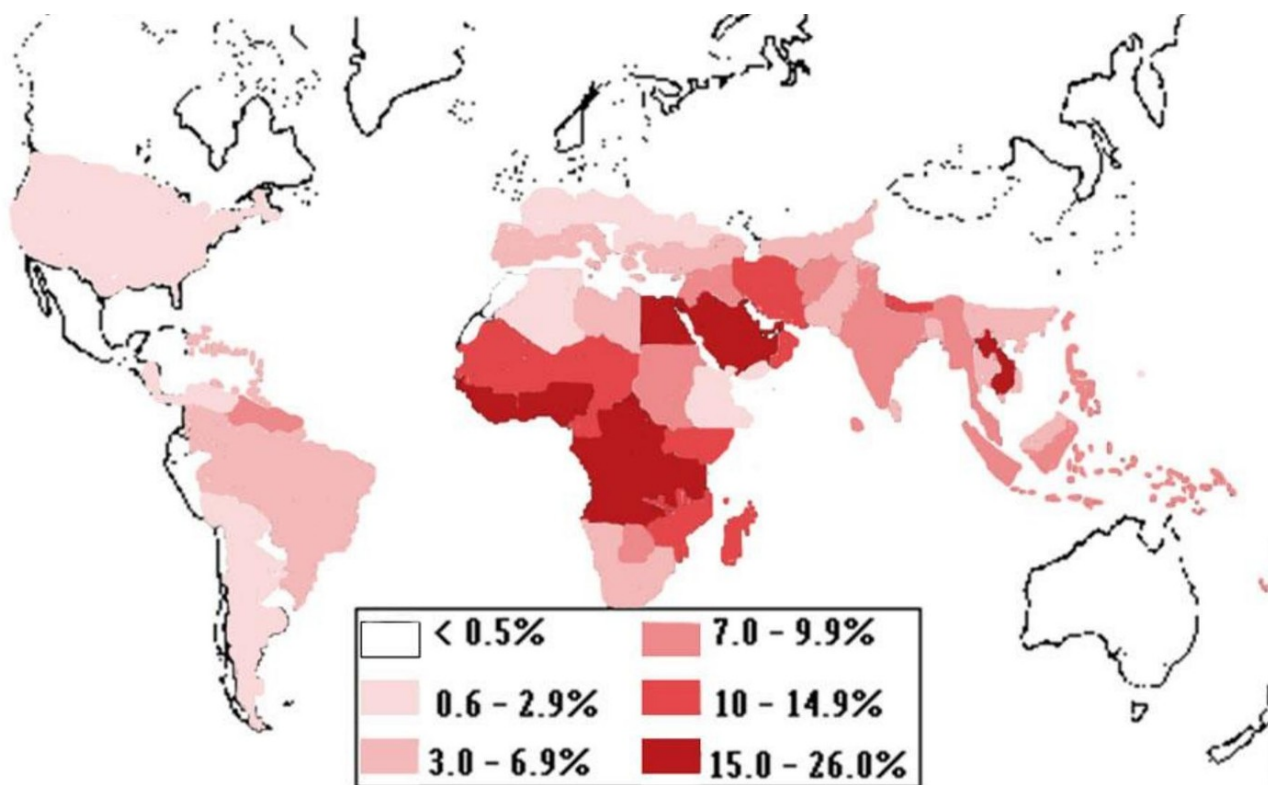


Figure HAEM-L1.08

Global distribution of G6PD deficiency (highest in Africa, Mediterranean, Middle East, S/SE Asia)

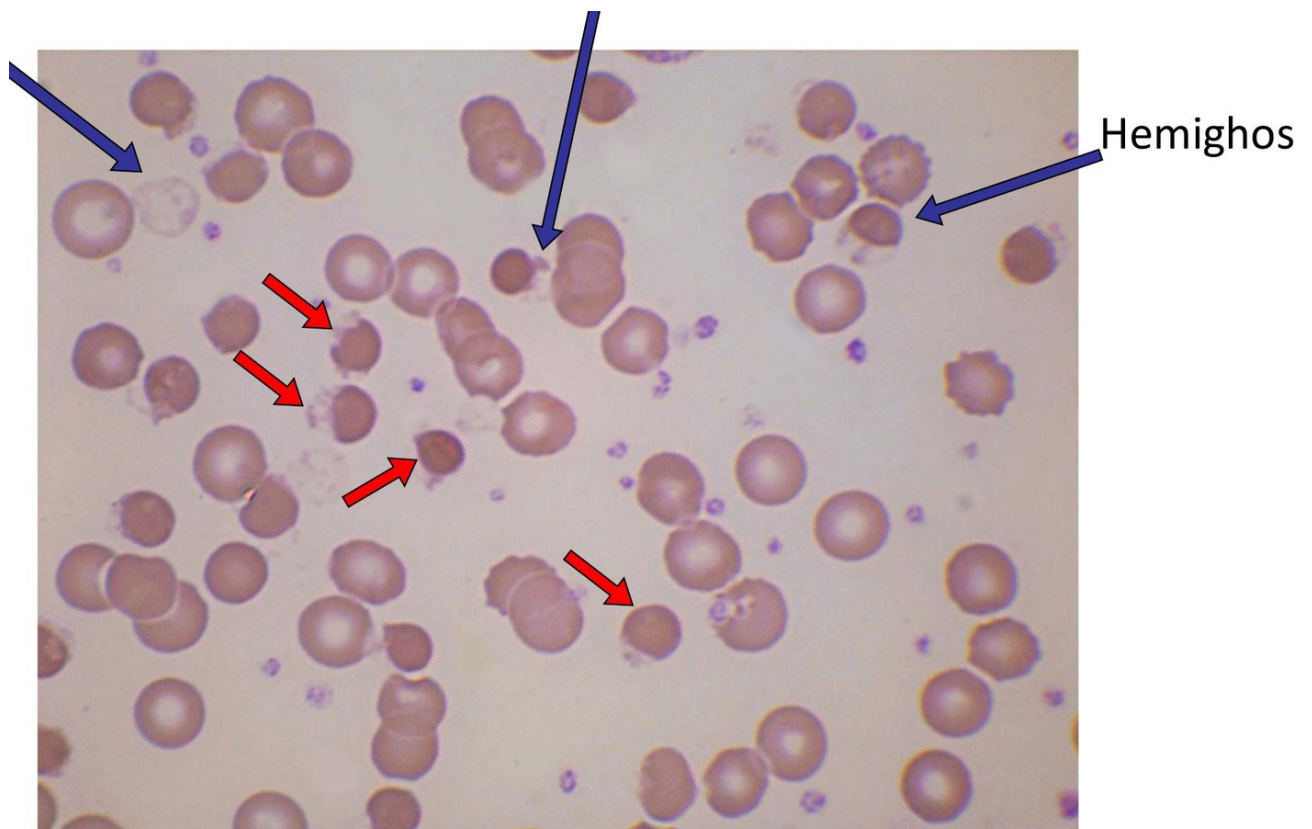


Figure HAEM-L1.09

G6PD haemolytic crisis: Heinz bodies, ghost cells, hemighosts, irregularly contracted cells

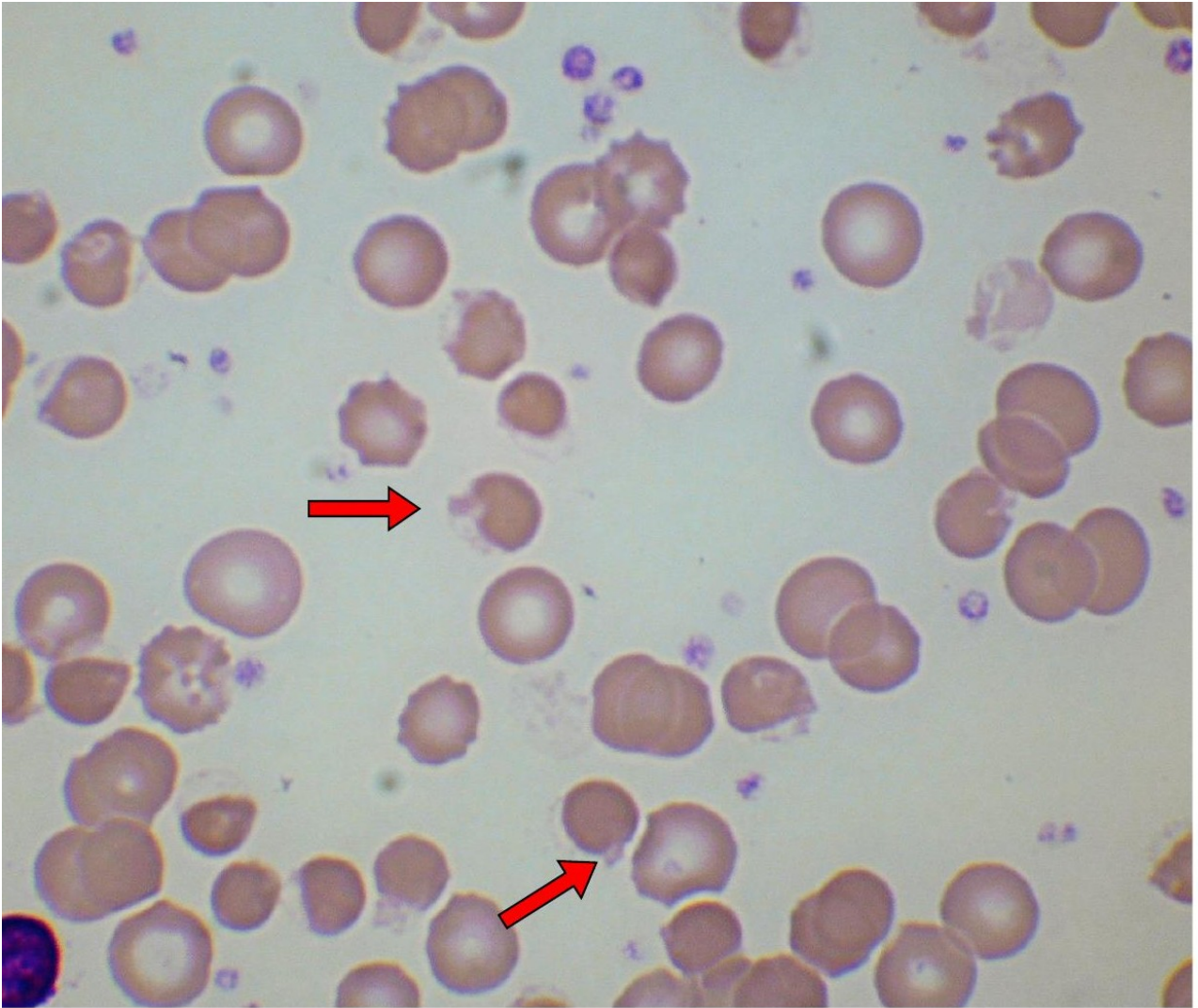
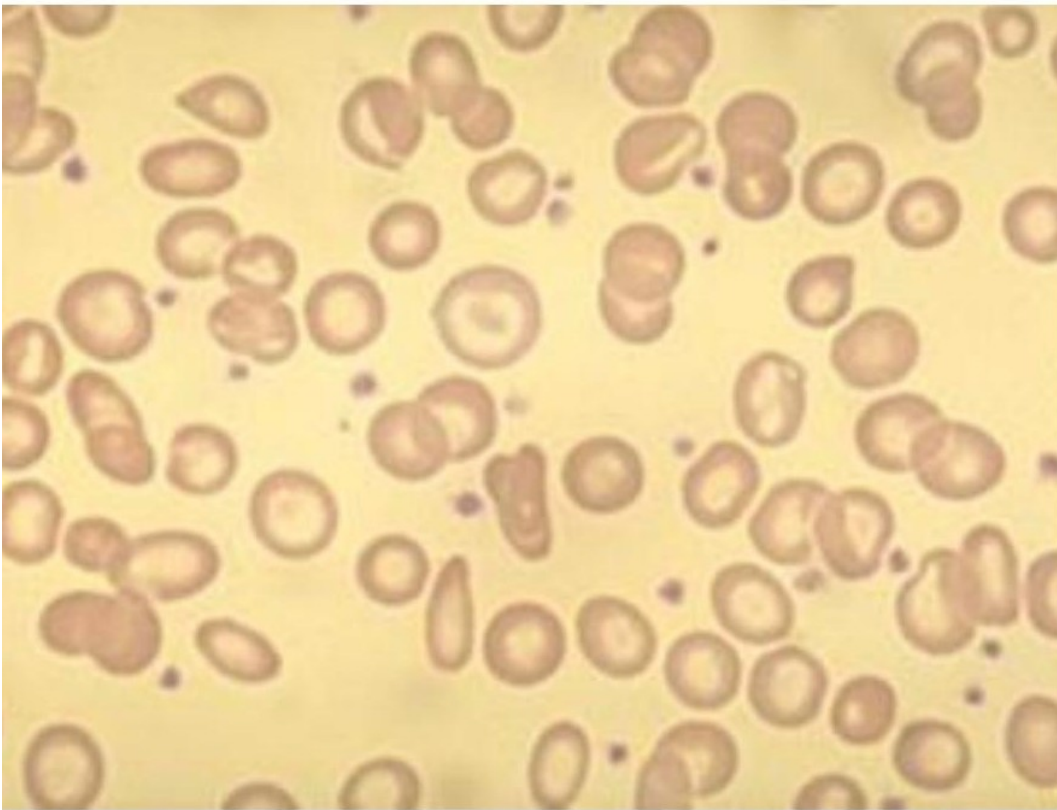


Figure HAEM-L1.10

Heinz bodies visible on standard Romanowsky stain

Blood film



Tran

Tran
(%)

Hb

Figure HAEM-L1.11

IDA blood film: microcytic, hypochromic, pencil cells (elliptocytes)

n – e.g. aspirin/non-steroidal anti-inflammatory drugs

Koilonychia



Glossitis
Angular stomatitis



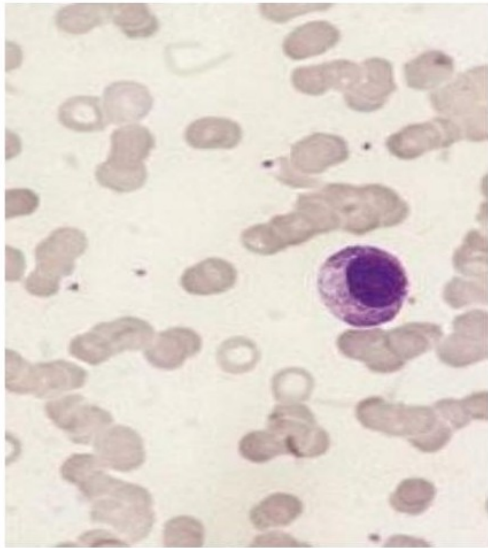
Summarise the pathology and pathophysiology of haematological disorders.

Describe the clinical features and treatment options of haematological disorders

Figure HAEM-L1.12

IDA clinical signs: koilonychia, glossitis, angular stomatitis

the most likely



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ders.

Figure HAEM-L1.13

Rouleaux ("stack of coins") — RBCs stacked from raised plasma proteins in inflammation



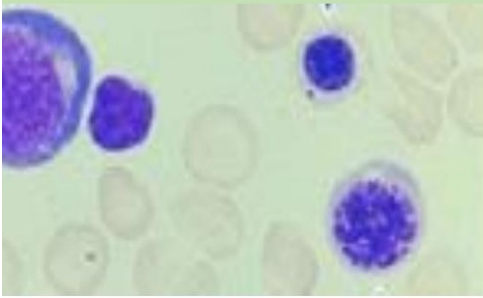
hematological disorders

Figure HAEM-L1.14

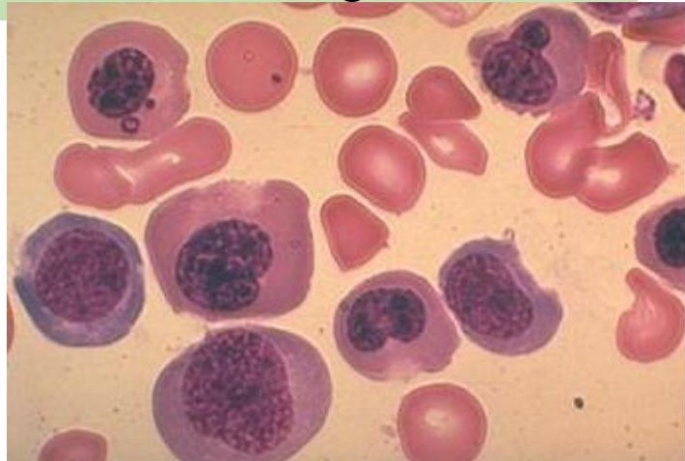
Megaloblastic blood film: oval macrocyte and hypersegmented neutrophil (arrowed)

impaired DNA synthesis, nuclear maturation and cell division

Adequate cytoplasmic maturation and cell growth



Normal BM

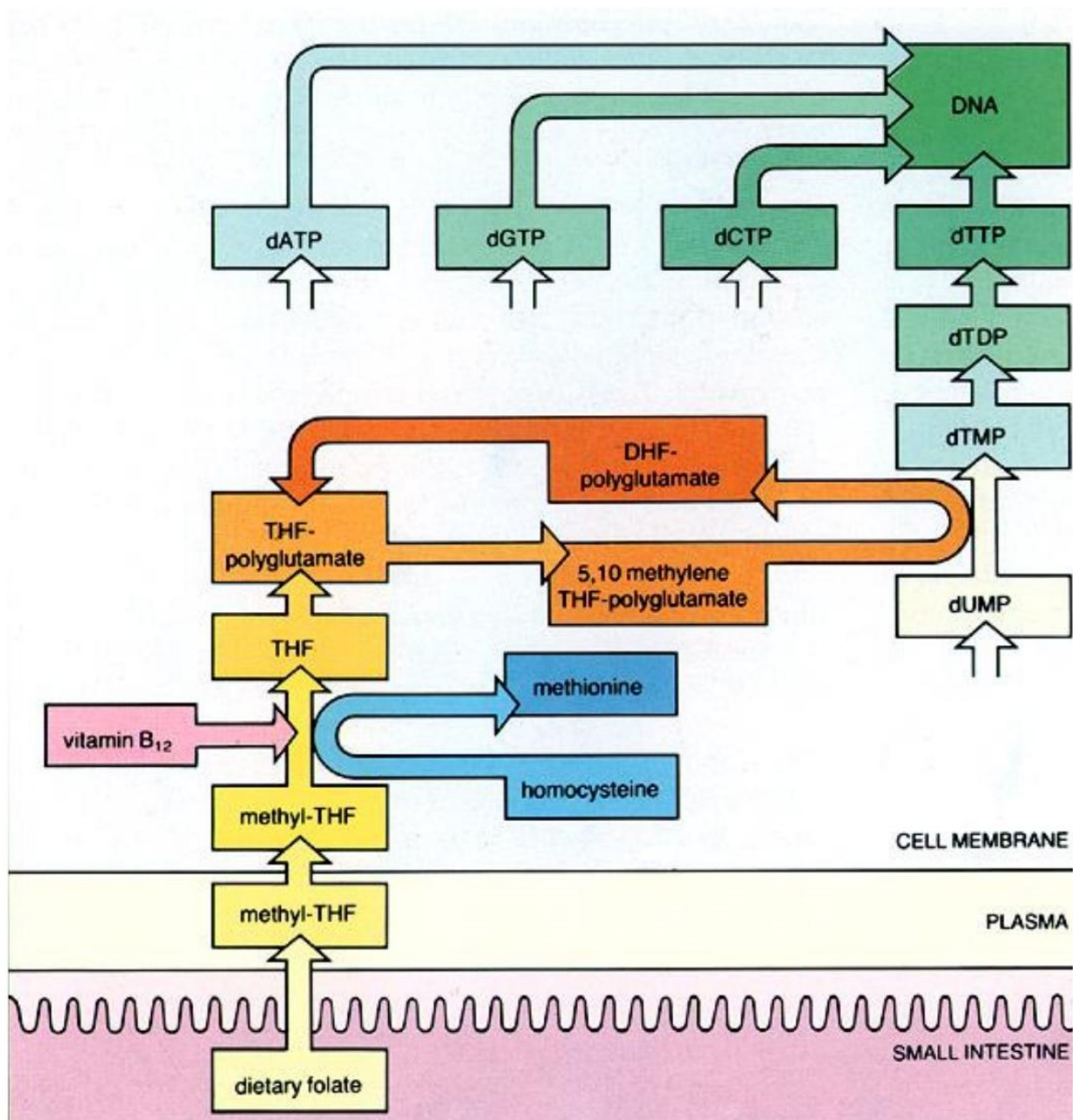


Megaloblastic BM

IR-10 Haematological disorders: Summarise the pathology and pathophysiology of haematological disorders

Figure HAEM-L1.15

Normal vs megaloblastic bone marrow: large precursors with asynchronous nucleocytoplasmic maturation



S-CVR-10 Haematological disorders: Summarise the pathology and pathophysiology of h:

Figure HAEM-L1.16

Folate-B12 biochemistry: dietary folate to methyl-THF to (B12 cofactor) to THF to DNA precursors

All these patients would be suspected of having a haemolytic anaemia—why?

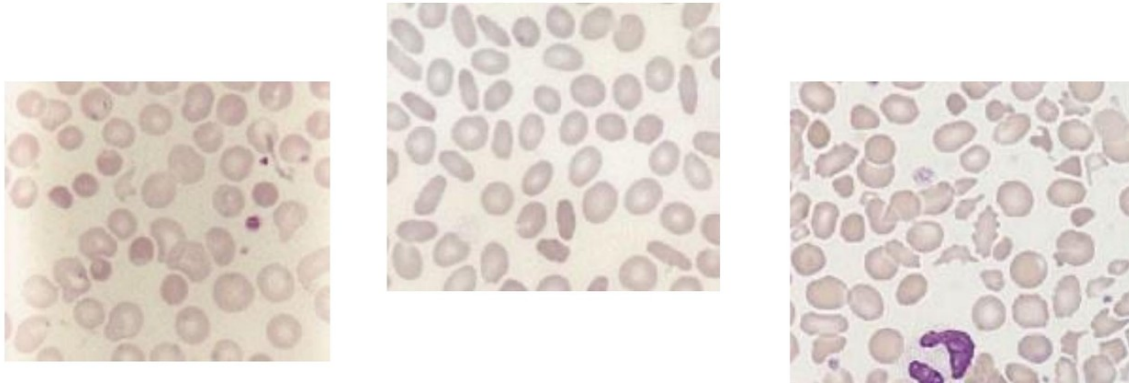


Figure HAEM-L1.17

Suspecting haemolytic anaemia: morphological clues on three blood films

Both these patients would also be suspected of having a haemolytic anaemia—why?

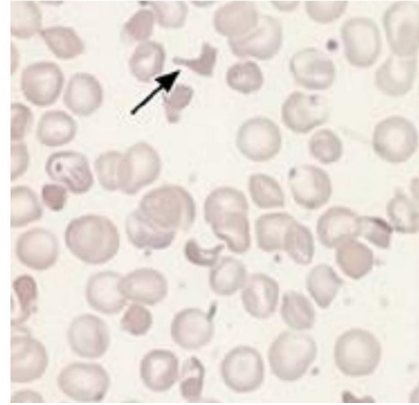
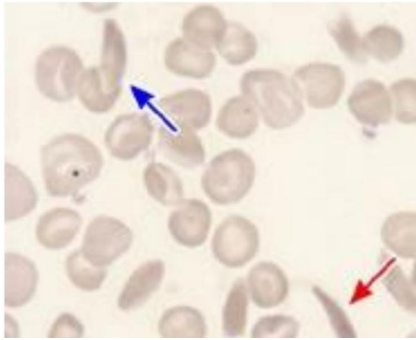


Figure HAEM-L1.18

Sickle cells (left) and red cell fragments / schistocytes (right)

These children would be suspected of having a haemolytic anaemia—why?



Figure HAEM-L1.19

Clinical signs in chronic haemolysis: pigment gallstones (cholangiogram) + scleral icterus

Classification of anaemia on the basis of cell size



Microcyte

cell that is **smaller** than normal



Normal red cell



Macrocyte

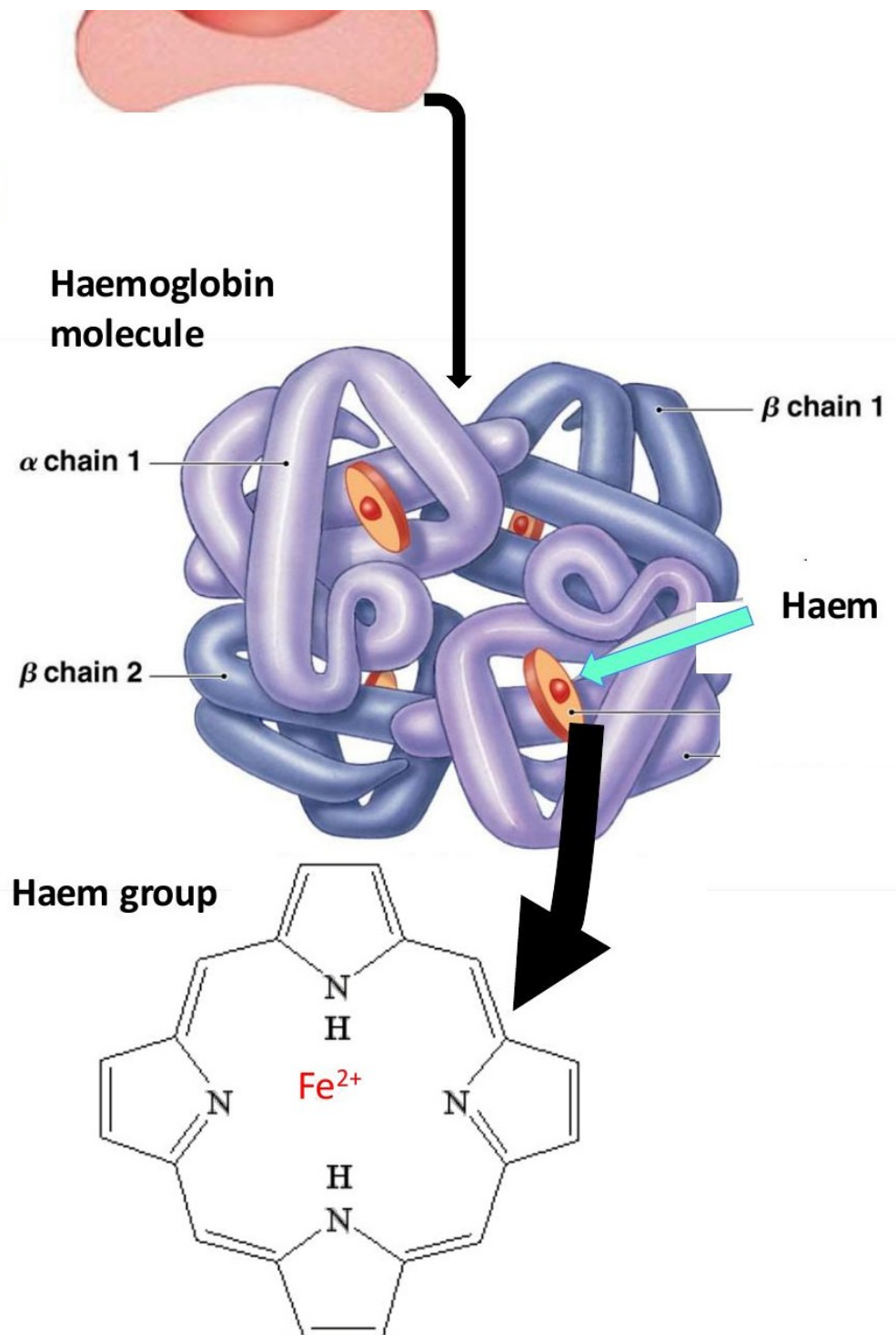
a red cell that is **larger** than normal

Figure HAEM-L1.20

Microcyte vs normal RBC vs macrocyte (single-cell view)

globin

SE



ders.

Figure HAEM-L1.21

Haemoglobin: 2 alpha + 2 beta globin chains, each binding one haem (Fe^{2+})